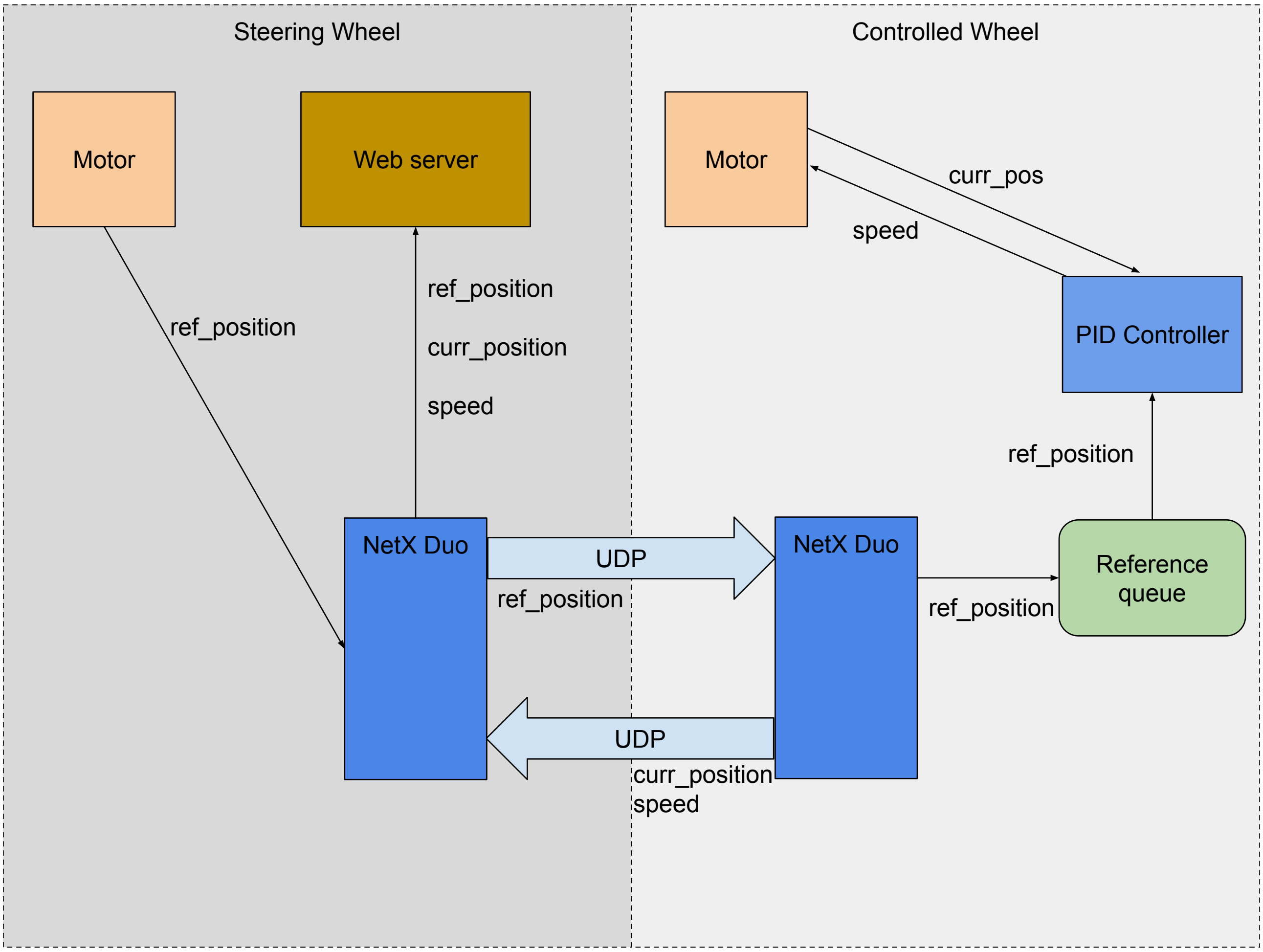
Project Documentation

This document describes the main functionality, information flow and core functions of the Project.

1 Overview

The system implements a digital motor controller by connecting two NUCLEO-H753ZI boards (A and B), each extended with a PWM controlled motor. Motor A acts as a steering wheel, Motor B as a controlled wheel. The two boards communicate data via UDP. Board A further acts as a webserver, displaying live graphs of the two motors.

2 Data Flow



3 Global Functions

a) Steering wheel

encoder_driver_initialize()

Initializes the motor driver by creating a mutex to protect position data on read/write operations. Must be called once to enable use of encoder_driver_get_position().

encoder_driver_get_position()

returns current motor position as a value between -256 and +256, where -256 and +256 correspond to -180° and +180° respectively.

nx_app_thread_entry(ULONG thread_input)

This is the core thread function of the program. It sends the reference position to the controlled wheel and receives current position and speed information back.

Parameter <u>thread_input</u>: unused.

b) Controlled wheel

motor_init()

Initializes the motor driver by creating mutexes to protect speed and positon data on read/write operations. Must be called once to enable use of motor_set_speed() and motor_get_pos().

<u>Returns</u> 0 if successful and an Error Code otherwise.

motor_set_speed(int speed)

Parameter speed: PWM vaule of the motor speed. Must be a value between -500 and +500. The sign of the value encodes the rotation direction.

<u>Returns</u> 0 if successful, Error Code (Mutex error) otherwise.

motor_get_pos(uint32_t* pos)

Parameter *pos: pointer to variable where current motor position should be stored. Will be a value between 0 and MOTOR_MAX_POS (256).

<u>Returns</u> 0 if successful, Error Code (Mutex error) otherwise.

controller_thread_entry(ULONG constants)

This is the function that controls the controller speed to track a provided reference position. It implements a PID Controller.

Parameter <u>constants</u>: Legacy, unused in the current state of the program.

nx_app_thread_entry(ULONG thread_input)

This function is responsible for receiving and sending UDP messages to the Steering Wheel. See 2. Data flow for details.

Parameter <u>thread_input</u>: unused.